Sardar Vallabhbhai National Institute of Technology, Surat

Unleashing the Real-World Applications of Artificial Intelligence and Machine Learning

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Explainable AI (XAI): Making AI Understandable to Humans

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AI Successes

- **Science:** Astronomy, neuroscience, medical imaging, bio-informatics
- **Environment:** Energy, climate, weather, resource
- **Retail:** Intelligent stock control, demographic store placement
- **Manufacturing:** Intelligent control, automated monitoring, detection methods
- **Security:** Intelligent smoke alarms, fraud detection
- **Marketing Promotions**
- **Management scheduling timetabling**
- **Finance Credit scoring and risk analysis**
- **Web data**: Information retrieval information extraction

Futuristic AI

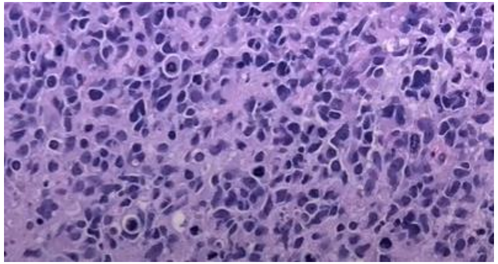
Futuristic cityscape with advanced technologies, symbolizing innovation and harmony between humans and AI.

Futuristic laboratory filled with advanced equipment, holographic displays, and robotic systems, symbolizing innovation and exploration.

Futuristic landscape blending technology and nature

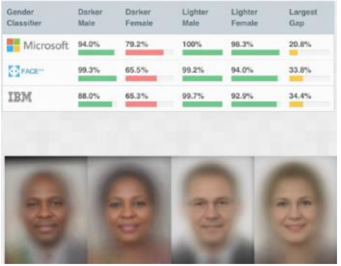
Where AI is yet to fulfill its promise

Arizona police released photographs from the pedestrian death involving an Uber self-driving car.



Disease misclassification

safety-critical systems: glass cockpit of a C-141, Space Shuttle and control room of a nuclear power plant.



Gender Classifier	Darker Male	Darker Female	Lighter Male	Lighter Female	Largest Gap
Microsoft	94.0%	79.2%	100%	98.3%	20.8%
FACE++	99.3%	65.5%	99.2%	94.0%	33.8%
IBM	88.0%	65.3%	99.7%	92.9%	34.4%

COTS gender classification

Characterizing these applications

- **Wrong decisions can be costly and dangerous**
- **Accuracy is not the only objective**
- **Need for multi-dimensional perspective**

Challenges and limitations of AI

Lack of Explainability: Many AI models, especially deep learning models, are complex and opaque. It can be difficult to understand how they arrived at a particular decision, making it challenging to trust and debug their outputs. This is particularly crucial in critical applications like healthcare or finance.

Ethical Dilemmas: Ethical concerns surrounding AI, such as bias, fairness, privacy, and autonomous decision-making.

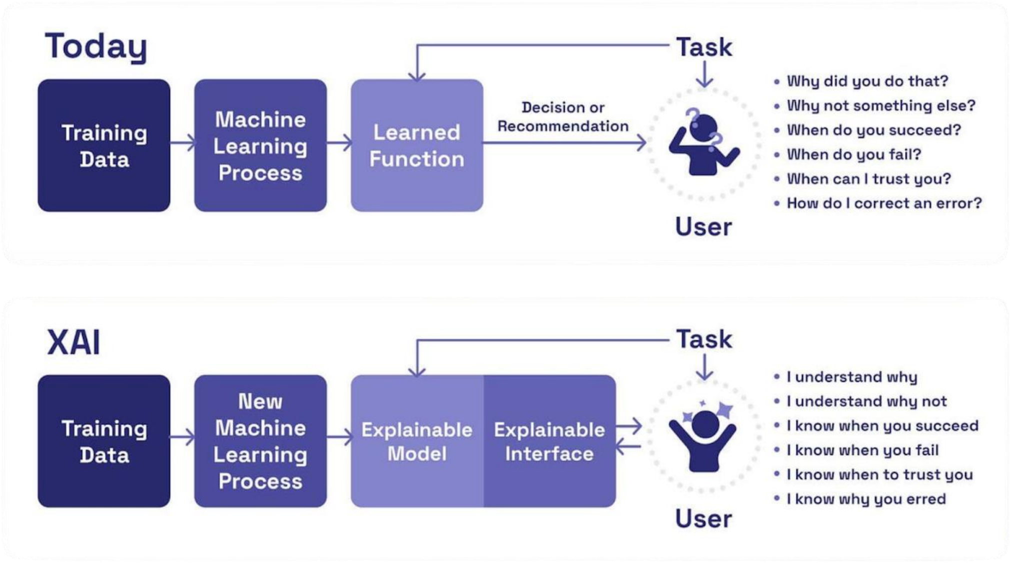
Robustness and Security: AI models can be vulnerable to adversarial attacks, where inputs are subtly modified to deceive the model into making incorrect predictions.

Interpretability: Understanding why an AI model makes specific decisions is critical in many applications. Lack of interpretability can hinder trust and adoption.

Regulatory Framework: A comprehensive and adaptive regulatory framework is needed to ensure the safe and responsible development and deployment of AI.

AI has the potential to revolutionize society, it is essential to address its limitations and ensure its ethical development to harness its full benefits.

Why Explainable AI ?



- **Explainable Model:** Develop a range of new or modified machine learning techniques to produce more explainable models.
- **Explainable Interface:** Integrate state-of-the-art HCI with new principles, strategies, and techniques to generate effective explanations.
- **Psychology of explanation:** summarize, extend, and apply current psychological theories of explanation to develop a computational theory.

European Union's General Data Protection Regulation

- Human-understandable rationale in decision-making
- Trust/confidence in system
- Compliance with ethical principles
- Enhanced control and robustness
- Openness of discovery and scientific research

European Union's General Data Protection Regulation (GDPR)

"A business using personal data for automated processing-making must be able to explain how the system makes decisions, See Article 15 (1) (h) and Recital of GDPR"

American Data Privacy and Protection Act

Key Provisions

Consumer Rights: The bill grants consumers various rights regarding their personal data, including the right to access, correct, and delete their data. Consumers would also have the right to opt-out of the sale of their personal data and to limit certain uses of their data.

Data Minimization: The ADPPA emphasizes the principle of data minimization, which means that businesses should only collect and use the data that is necessary for their legitimate purposes.

Privacy by Design: The bill encourages businesses to adopt a privacy-by-design approach, which means incorporating privacy considerations into the development and design of their products and services.

Enforcement: The ADPPA would be enforced by the Federal Trade Commission (FTC), which would have the authority to investigate violations and impose penalties.

India's Digital Personal Data Protection Act

India's Digital Personal Data Protection Act (DPDP) , 2023, India

An Act to provide for the processing of digital personal data in a manner that recognises both the right of **individuals to protect their personal data** and the need to process such personal data for lawful purposes and for matters connected therewith or incidental thereto.

Right to an explanation, Right of Access, Right to Deletion, Right to Correction, Right to Refuse Data Transfers, and Right to Opt-Out of Targeted Advertising

Goal of XAI: FATE in AI

Fairness

Accountability

Transparency

Ethics

Fairness: Mitigating bias and discrimination in AI-driven decision-making to promote equitable outcomes.

Transparency: Ensuring that AI systems are transparent and their decision-making processes are understandable by users and stakeholders.

Accountability: Establishing clear lines of responsibility and liability for the actions and decisions of AI systems.

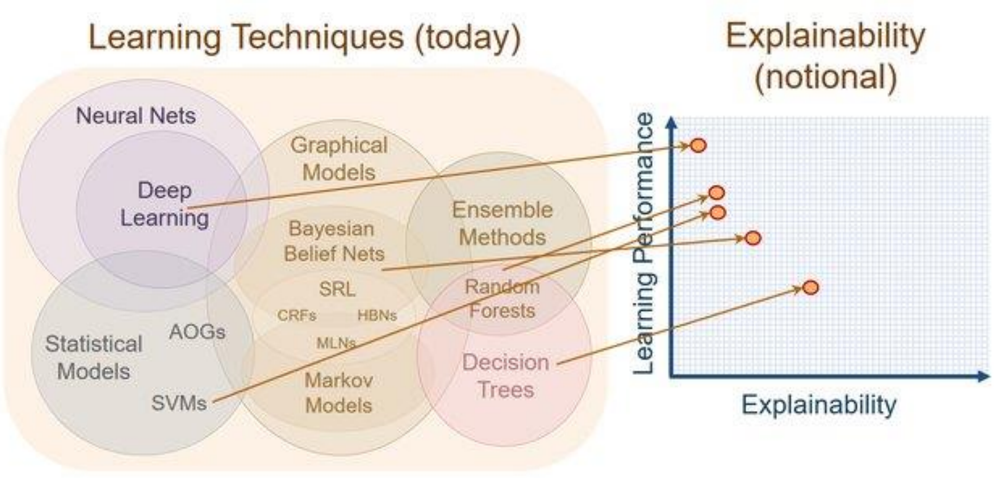
Privacy: Protecting sensitive data and personal information used in the development and deployment of XAI systems.

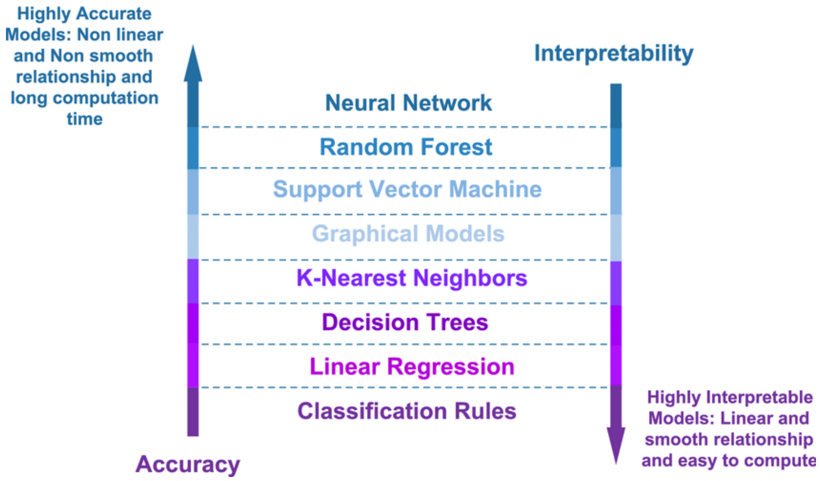
Artificial Intelligence (AI) is at the forefront of modern technology, and its effects are felt in many areas of society. To prevent algorithmic disparities, fairness, accountability, transparency, and ethics (FATE) in AI are being implemented

Credit: https://www.microsoft.com/en-us/research/theme/fate/

Credit: Lecue et al., Tutorial on XAI. AAAI 2020. https://xaitutorial2020.github.io/

Trade-off between accuracy vs. explainability



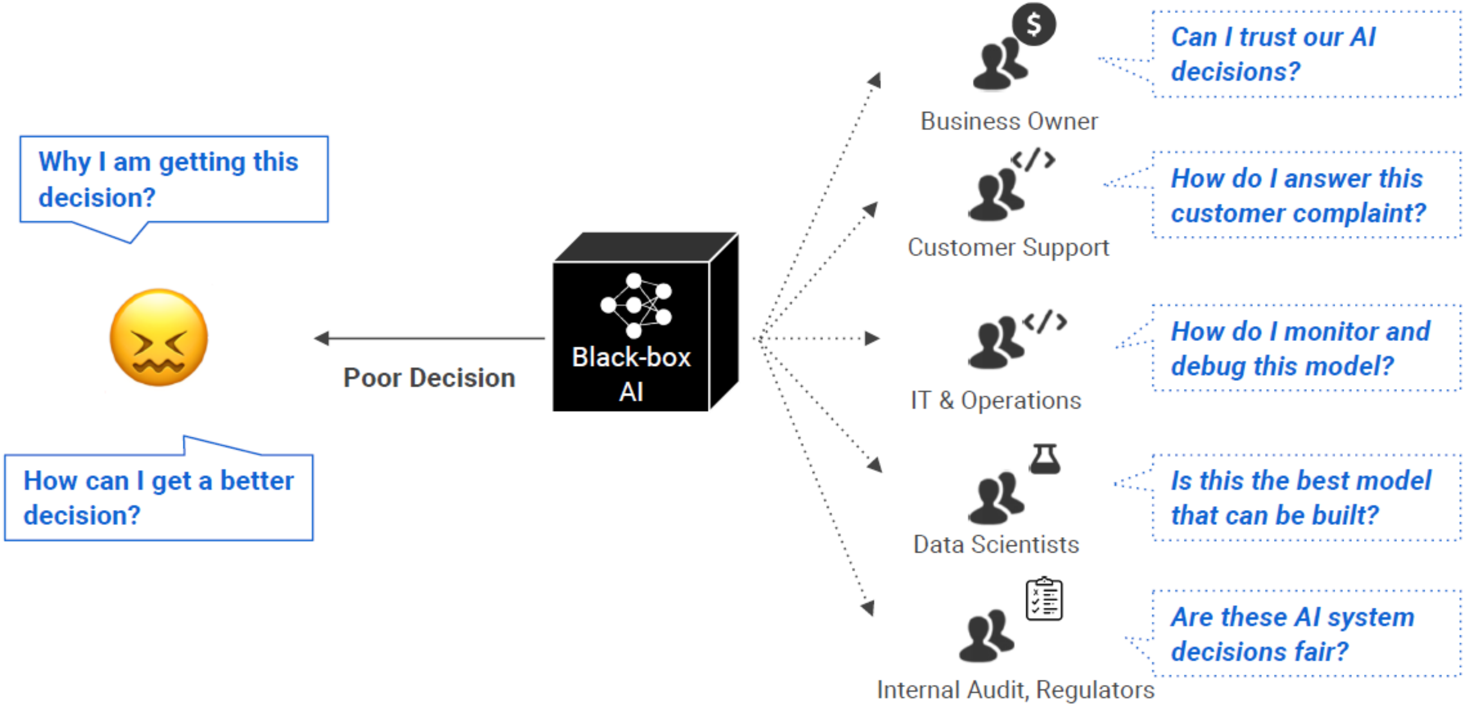


A trade-off between accuracy and explainability

67% of the businesses leaders taking part in PwC's 2017 Global CEO Survey believe that **AI and automation will impact negatively on stakeholder** trust levels in their industry in next five years.

Source: Taymouri et al., Business Process Variant Analysis: Survey and Classification

Whom is XAI for ?



Explainability vs. Interpretability

Interpretability

- Understand what a model did or might have done. For example, interpretability can help you understand what it means when a linear regression model calculates sums of weighted input features.

Explainability

- Summarizing the reason for neural network behaviour, gaining trust of users, producing insights or causes of decisions. For example, explainability can help you understand why a data point was sorted into a particular category

simple analogy to summarize the difference:

Interpretability is like having an owner's manual for a car. You understand how the engine, brakes, and steering work together.
Explainability is like having a mechanic tell you why your car broke down, even if you don't fully understand the inner workings of an engine yourself. They can point to a faulty part and explain its impact.

XAI Taxonomy

Agnosticity

Model-**agnostic**	Applicable to all model types
Model-**specific**	Only applicable to a specific model type

Scope

Global explanation	Explaining the whole model
Local explanation	Explaining individual predictions

Data Type

- Graph
- Image
- Text
- Tabular

Explanation Type

Visual

Data visualisation techniques may be used to understand the prediction or choice made over the input data.

Feature importance

After all possible combinations, we get the feature importance based on its average predicted marginal contribution to the model's decision.

Data points

This category includes all methods that return data points (already existent or newly created) to make a model interpretable.

Surrogate models

We can explain our complex model's prediction by using a simplified model (surrogate model) to approximate it around the prediction.

LIME: Local Interpretable Model agnostic Explanation

"Why Should I Trust You?"
Explaining the Predictions of Any Classifier

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ABSTRACT

Despite widespread adoption, machine learning models remain mostly black boxes. Understanding the reasons behind predictions is, however, quite important in assessing *trust*, which is fundamental if one plans to take action based on a prediction, or when choosing whether to deploy a new model. Such understanding also provides insights into the model, which can be used to transform an untrustworthy model or prediction into a trustworthy one.

In this work, we propose LIME, a novel explanation technique that explains the predictions of *any* classifier in an interpretable and faithful manner, by learning an interpretable model locally around the prediction. We also propose a method to explain models by presenting representative individual predictions and their explanations in a non-redundant way, framing the task as a submodular optimization problem. We demonstrate the flexibility of these methods by explaining different models for text (e.g. random forests) and image classification (e.g. neural networks). We show the

how much the human understands a model's behaviour, as opposed to seeing it as a black box.

Determining trust in individual predictions is an important problem when the model is used for decision making. When using machine learning for medical diagnosis [6] or terrorism detection, for example, predictions cannot be acted upon on blind faith, as the consequences may be catastrophic.

Apart from trusting individual predictions, there is also a need to evaluate the model as a whole before deploying it "in the wild". To make this decision, users need to be confident that the model will perform well on real-world data, according to the metrics of interest. Currently, models are evaluated using accuracy metrics on an available validation dataset. However, real-world data is often significantly different, and further, the evaluation metric may not be indicative of the product's goal. Inspecting individual predictions and their explanations is a worthwhile solution, in addition to such metrics. In this case, it is important to aid users by suggesting which instances to inspect, especially for large datasets.

LIME: Local Interpretable Model Agnostic Explanation

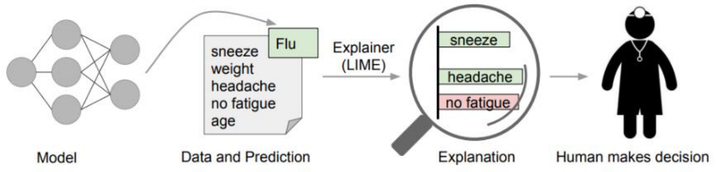
Local**:** Instead of explaining the predictions globally by building a global surrogate model, LIME focuses on training local surrogate models to explain individual prediction.

Interpretable**:** The Idea behind LIME is to make easy to interpretable local model such as linear regression and classification which can explain your black box model locally.

Model Agnostic: LIME can be applied to any black box model irrespective of the technique as long as it can predict probability .

Explanations**:** LIME will be able to explain how the model behaves, which features it picks up and what kinds of interactions between them takes place to drive the predictions

- Lime is a post hoc technique, which means that this is applied **after the event i.e. after model training.**
- Model internals are “hidden”, it works on tabular, image, graph, and text data.
- Explanations are **locally** faithful, but not necessarily globally

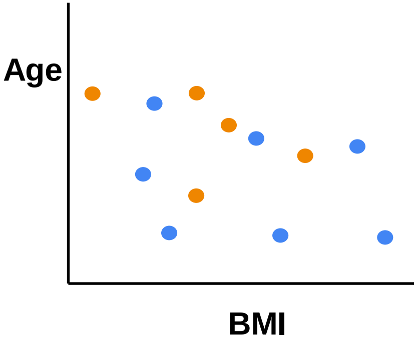


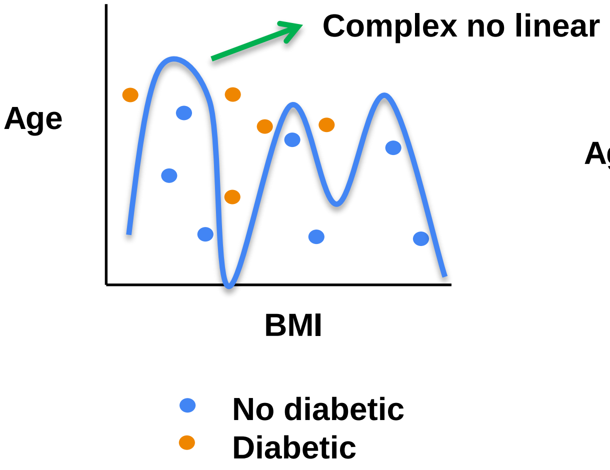
I have a medical emergency. Hence won’t be able to attend the meeting today.	Important
Hi may I get the information about your service?	Not important

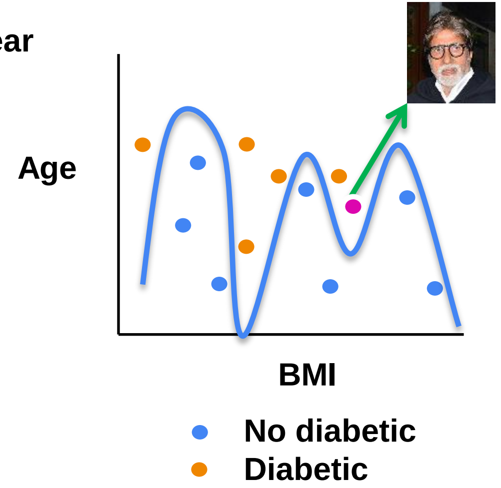
I have a **medical emergency**. Hence **won’t** be able to **attend** the **meeting** today.	Important
Hi may I get the **information** about your **service**?	Not important

LIME: Diabetic Database

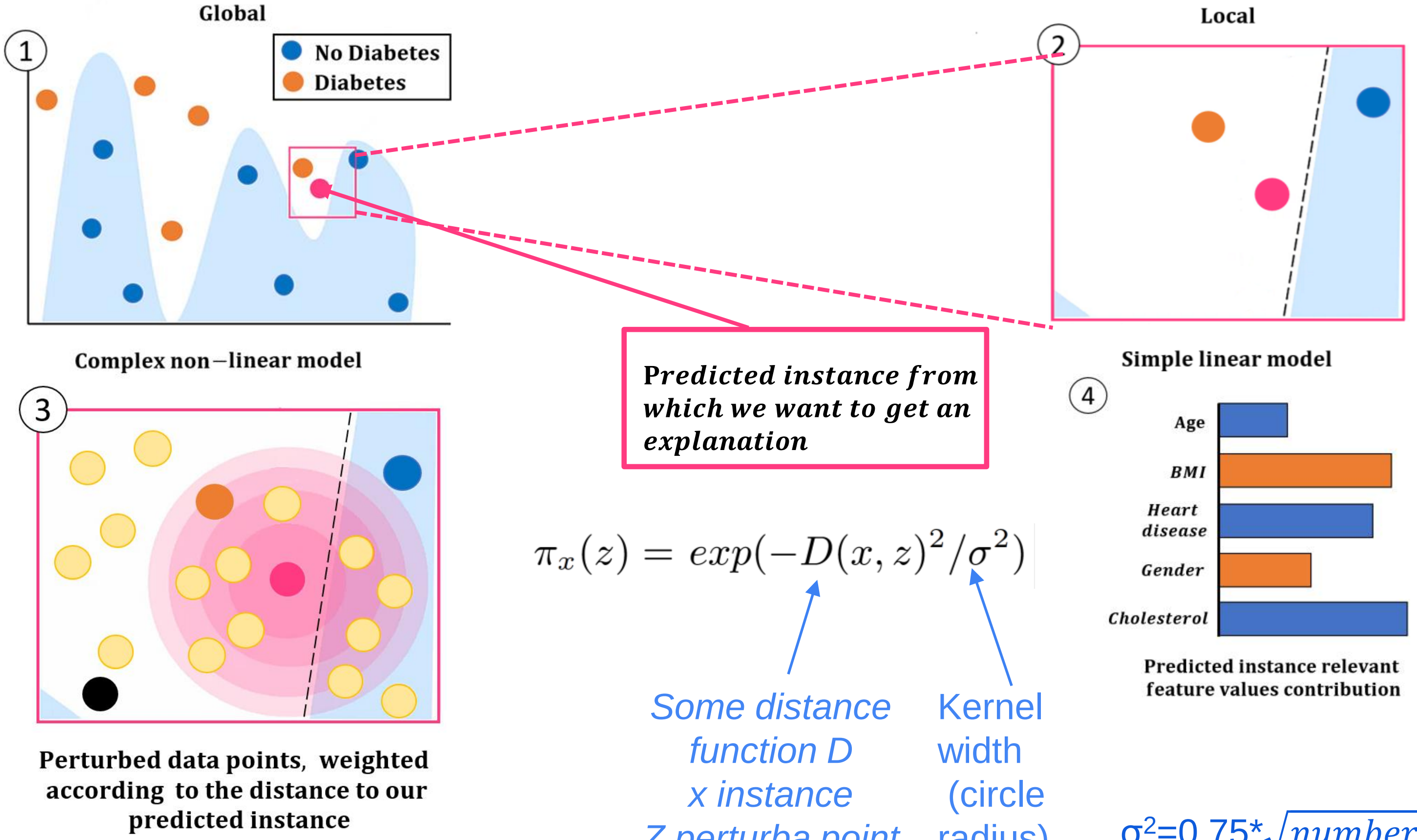
Pregnancies	Glucose	Blood Pressure	Skin Thickness	Insulin	BMI	Diabetes Pedigree Function	Age	Outcome
6	148	72	35	0	33.6	0.627	50	**1**
1	85	66	29	0	26.6	0.351	31	**0**
8	183	64	0	0	23.3	0.672	32	**1**
1	89	66	23	94	28.1	0.167	21	**0**







LIME step by step



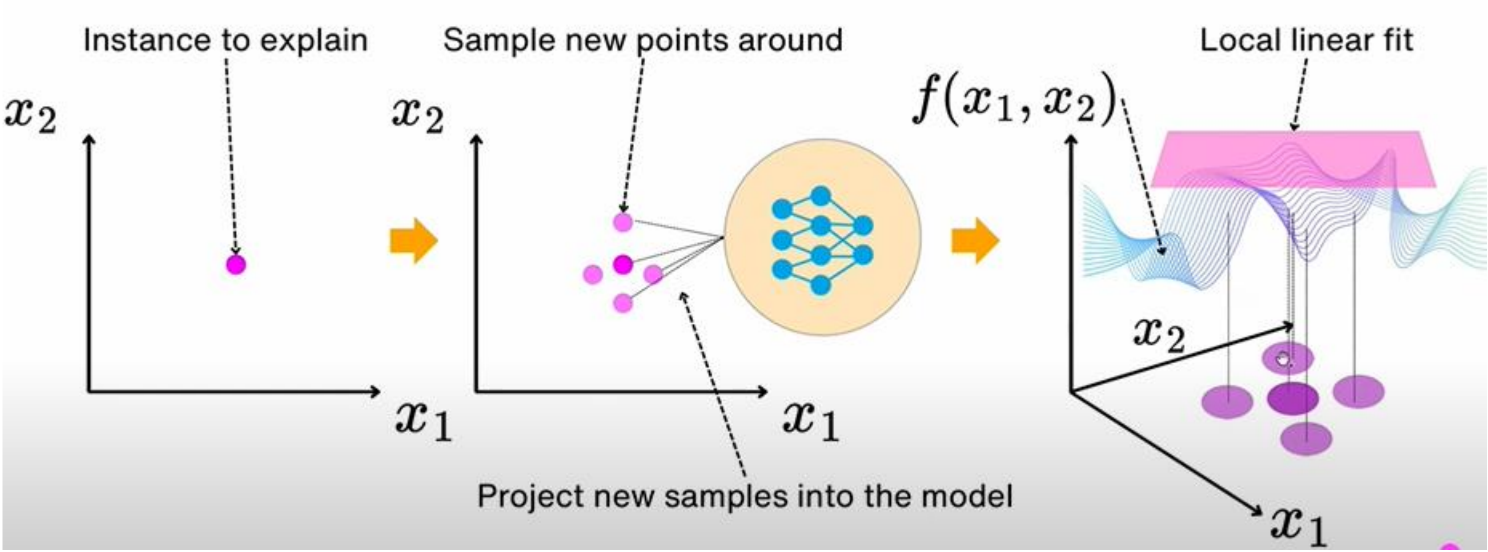
$$\pi_x(z) = exp(-D(x,z)^2/\sigma^2)$$

Some distance function D
x instance
Z perturba point

Kernel width (circle radius)

$\sigma^2=0.75*\sqrt{number\ of\ features}$

LIME with tabular data



$f(x_1,x_2)$ is the global model (complex-linear model)

The math in LIME

$$\xi(x) = \underset{g \in G}{\arg\min}\ \boxed{\mathcal{L}(f, g, \pi_x)} + \boxed{\Omega(g)}$$

Zeta

Family of interpretable models (Linear regression)

Complex model being explained

$f : \mathbb{R}^d \rightarrow \mathbb{R}$

Simple/surrogate interpretable model

Local neighbourhood of x (Proximity)

Complexity measure of g Regularizar

Predicted instance we want to explain

$x \in \mathbb{R}^d \longrightarrow$ number of features

$Age = 56$	$Gender = F$	$BMI = 30$	......	$diabetic = yes$

Loss terms

$$\xi(x) = \underset{g \in G}{\arg\min}\, \mathcal{L}(f, g, \pi_x) + \Omega(g)$$

Train a weighted, interpretable model on the dataset with the perturbed instances

f: The complex "black box" model (e.g., a deep neural network or XGBoost). g: The "interpretable" surrogate model (e.g., a sparse linear regression or decision tree).

x: The specific instance being explained. pi_x: The Proximity Kernel, which defines the size of the neighborhood around x. L(f, g, pi_x): The local fidelity loss. It measures how unfaithful g is to f in the locality defined by pi_x

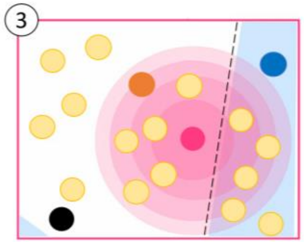
(1) $$\mathcal{L}(f, g, \pi_x) = \sum_{z \in \mathcal{Z}} \pi_x(z) \left(f(z) - g(z)\right)^2$$

Omega(g): The complexity penalty
(e.g., the number of non-zero weights in a linear model).

Complex model prediction

Simple model prediction

We want to know the local behaviour of *L in the vicinity of x. Then we generate perturbation $\pi_{x.}$ Z are the perturb points.*
$\pi_{x.}$ (z) is distance function between x(instance) and z(perturb).
When the value of π_x is less then point are very far .



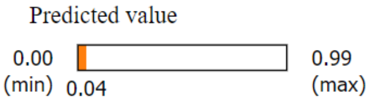
Perturbed data points, weighted according to the distance to our predicted instance

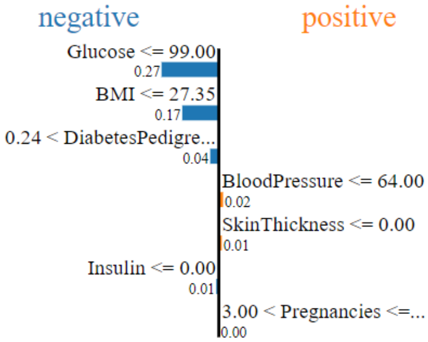
(2) $\Omega(g)$ LIME uses sparse linear models (K – LASSO)

$$J(\theta) = \underbrace{\frac{1}{2n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}_{\text{MSE (Loss)}} + \lambda \underbrace{\sum_{j=1}^{m}|\theta_j|}_{\text{L1 Penalty}}$$

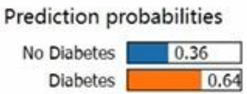
LIME Result on Diabetic Database

```
Intercept 0.5191701535536987
Prediction_local [0.0678266]
Right: 0.04
```





Feature	Value
Glucose	73.00
BMI	26.80
DiabetesPedigreeFunction	0.27
BloodPressure	60.00
SkinThickness	0.00
Insulin	0.00
Pregnancies	5.00



Feature	Value
Glucose	98.00
Age	43.00
BMI	34.00
BloodPressure	58.00
Insulin	190.00
SkinThickness	33.00
DiabetesPedigreeFunction	0.43
Pregnancies	6.00

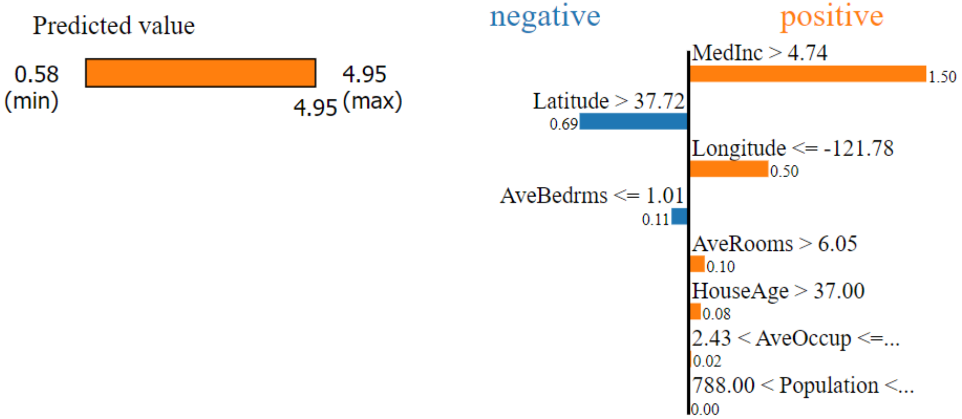
LIME application on tabular data

- *Housing public tabular database*
- *Random forest fit with this database*
- *LIME explanations*

	MedInc	HouseAge	AveRooms	AveBedrms	Population	AveOccup	Latitude	Longitude	label
0	8.3252	41.0	6.984127	1.023810	322.0	2.555556	37.88	-122.23	4.526
1	8.3014	21.0	6.238137	0.971880	2401.0	2.109842	37.86	-122.22	3.585
2	7.2574	52.0	8.288136	1.073446	496.0	2.802260	37.85	-122.24	3.521
3	5.6431	52.0	5.817352	1.073059	558.0	2.547945	37.85	-122.25	3.413
4	3.8462	52.0	6.281853	1.081081	565.0	2.181467	37.85	-122.25	3.422

LIME Result on Housing dataset

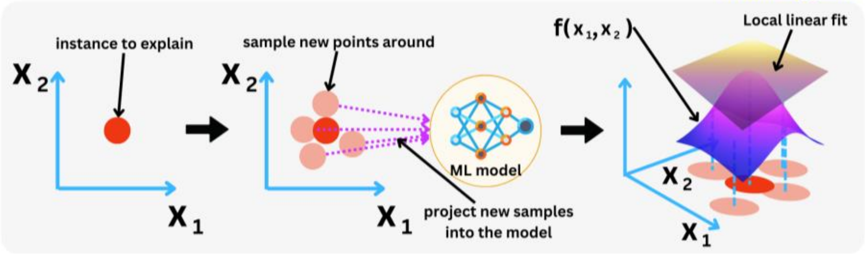
```
Intercept 1.9124218285307681
Prediction_local [3.31290903]
Right: 4.952499499999999
```



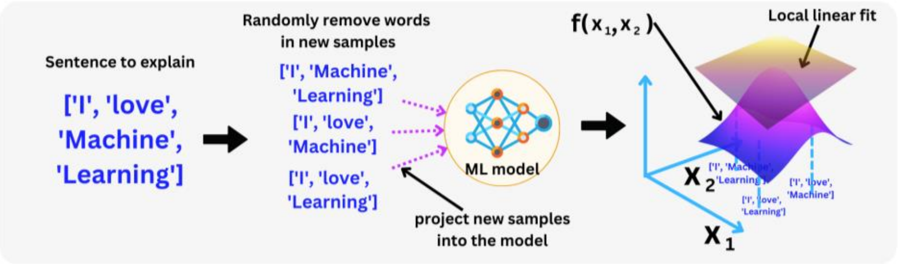
Feature	Value
MedInc	10.71
Latitude	37.79
Longitude	-122.49
AveBedrms	0.95
AveRooms	8.43
HouseAge	52.00
AveOccup	2.73
Population	805.00

- **Right**: prediction given by trees regressor prediction model.
- **Prediction_local:** This denotes the value outputted by a linear model trained on the perturbed samples.

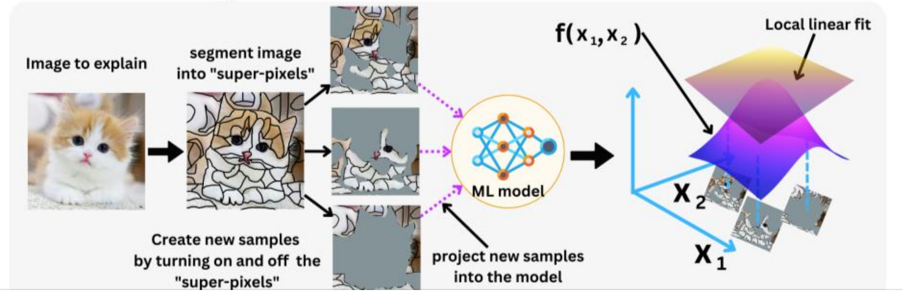
LIME with Tabular data



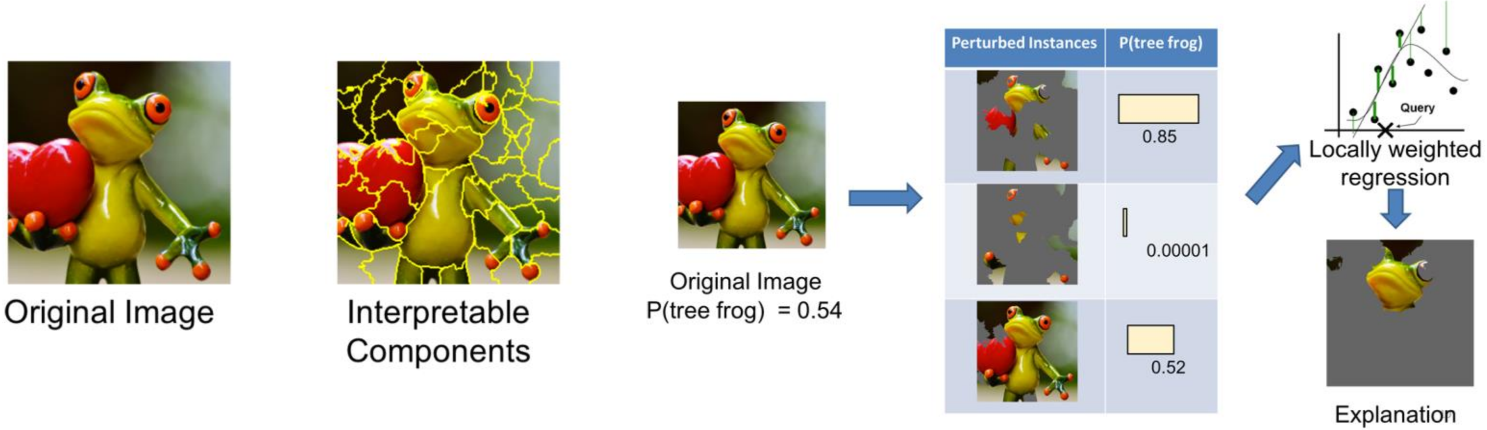
LIME with Text data



LIME with Image data



LIME works for image classification

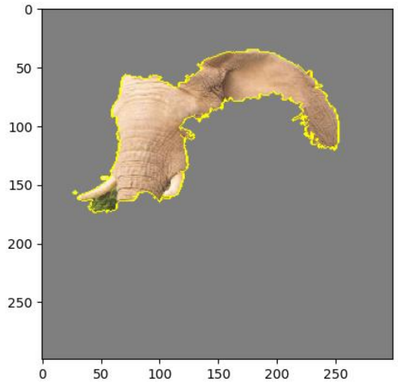


1. Generate a data set of perturbed instances by turning some of the interpretable components “off” (making them gray / means not contributing in classification).
2. For each perturbed instance, we get the probability that a tree frog is in the image according to the model.
3. Learn a simple (linear) model on this data set, which is locally weighted—that is, we care more about making mistakes in perturbed instances that are more similar to the original image.
4. Present the superpixels with highest positive weights as an explanation, graying out everything else.

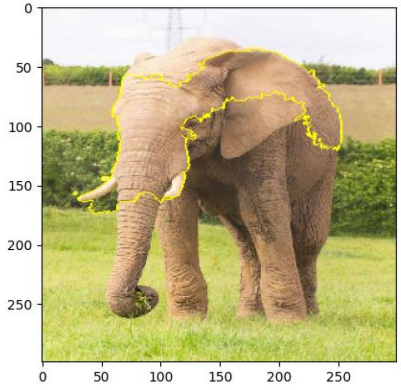
Popular code: https://github.com/marcotcr/lime

LIME application on image

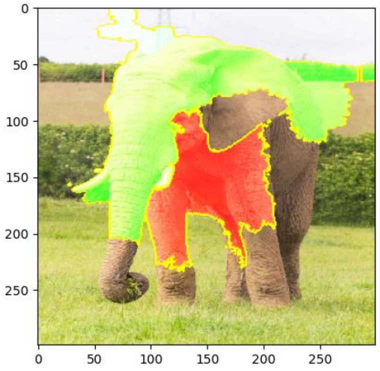
Input image



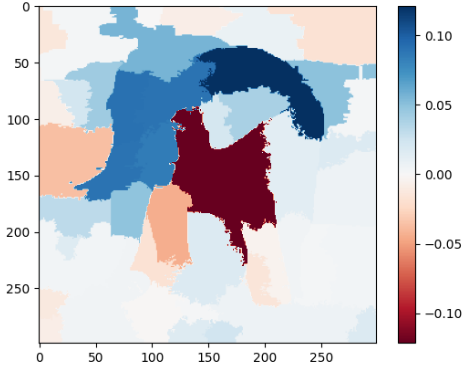
Superpixel for the top most Prediction



Superpixel for the top most Prediction



Positive and negative



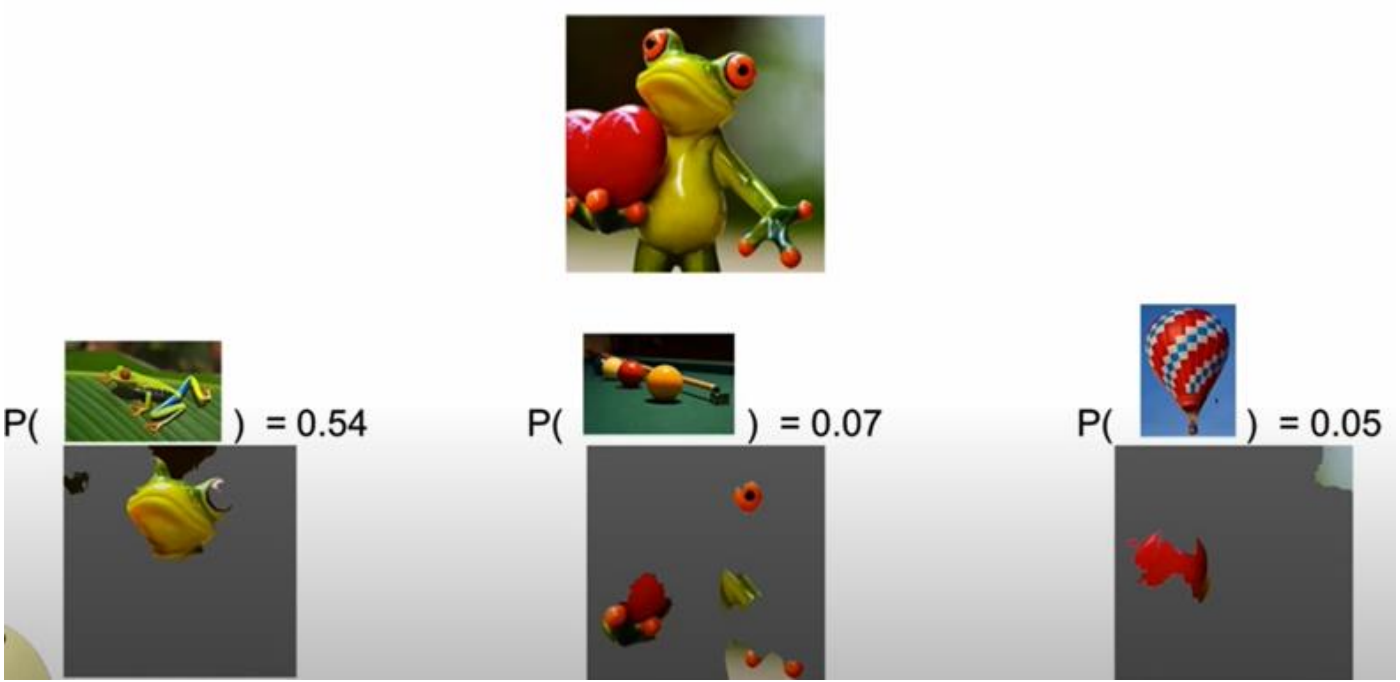
Heat map

https://github.com/prodramp/DeepWorks/tree/main/MLI-XAI

https://www.youtube.com/watch?v=dQ_jvRkzN1Q&t=108s

https://www.youtube.com/watch?v=d6j6bofhj2M

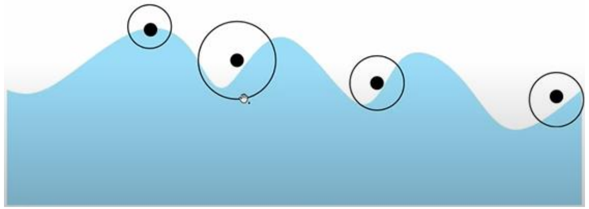
What does the neural network see?



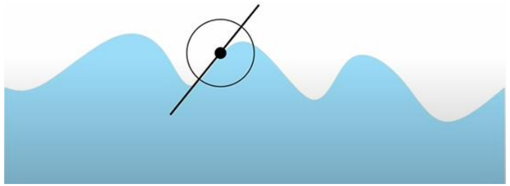
Limitation of LIME

Local vs. Global Interpretability:

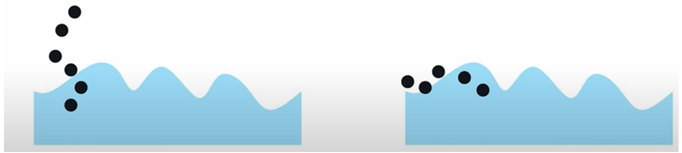
- LIME explains individual predictions, which might not provide the comprehensive insights into the model's overall behaviour.
- How to defined the size of circle.



- **Linear Assumption:** LIME fits a linear model to approximate the original model's decision making, which might oversimplify or inaccurately represent the complex decision boundaries.



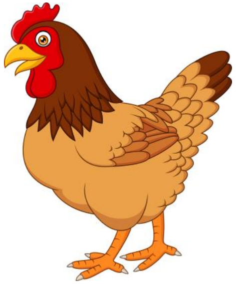
- **Stability and consistency:** LIME's random sampling can lead to different explanations for the same input, raising concerns about reliability



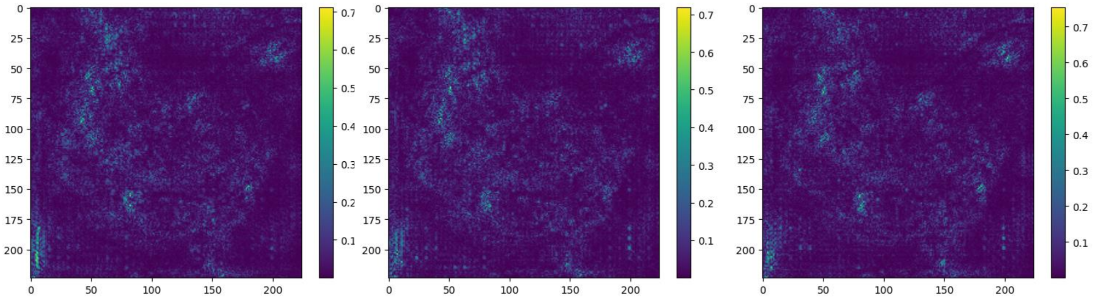
Saliency Map (Which pixels are more important)

- A **Saliency Map** in the context of a Convolutional Neural Network (CNN) is a visualization technique that highlights the regions of an input image that contribute most to the model's prediction.
- Saliency maps is a technique to rank the pixels in an image based on their contribution to the final score from a Convolution Neural Network.

$$\boldsymbol{Backpropagation = \frac{\partial F(image)}{\partial (w)}}$$



$$\boldsymbol{Saliency\ map = \frac{\partial F(image)}{\partial (image)}}$$



Simonyan et. al Deep Inside Convolutional Networks: Visualising Image Classification Models and Saliency Maps

Saliency Map

1. Input Image and Forward Pass:

Pass the input image X through the CNN to obtain the class scores $\mathbf{S_c}$ where **c** is the target class.

2. Select the Target Output:

Choose the output score $\mathbf{S_c}$ corresponding to the class of interest.

3. Compute the Gradient:

Compute the gradient of the target class score $\mathbf{S_c}$ with respect to the input image

This gradient measures how sensitive the $\nabla_X S_c$ **c** is to small changes in the input image X

4. Backpropagation:

Use backpropagation to propagate the gradients from the output $\mathbf{S_c}$ back to the input image. This gives the pixel-wise importance for the target class.

5. Take the Absolute Value or Maximum Gradient:

Combine the gradients across the RGB (i,j,c) channels by taking the maximum absolute value across channels for each pixel:

$$M[i,j] = \max_{c \in \{R,G,B\}} |\nabla_X S_c[i,j,c]|$$

Alternatively, compute the norm of the gradient $M[i,j] = \sqrt{\sum_{c \in \{R,G,B\}} (\nabla_X S_c[i,j,c])^2}$

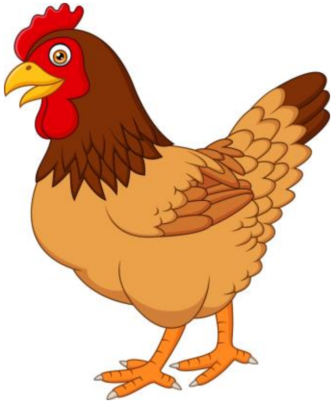
Saliency Map

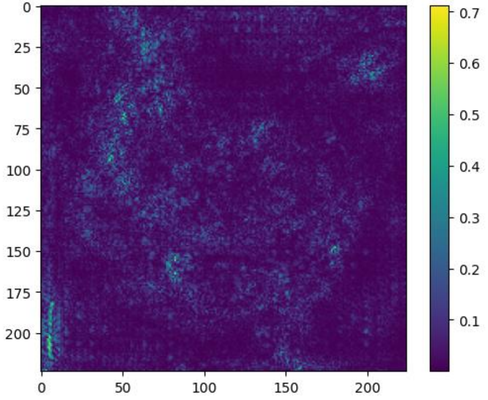
6. Normalize the Saliency Map:

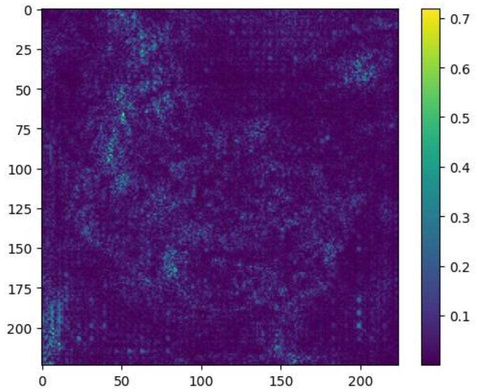
Normalize the saliency values to lie within a fixed range (e.g., 0 to 1) for visualization.

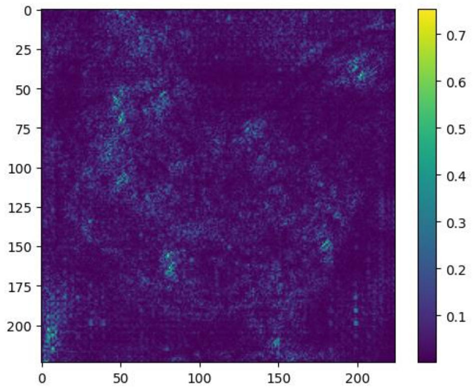
7. Visualize the Saliency Map:

Overlay the saliency map on the original image or display it as a heatmap to interpret the regions that are most important for the model's decision.





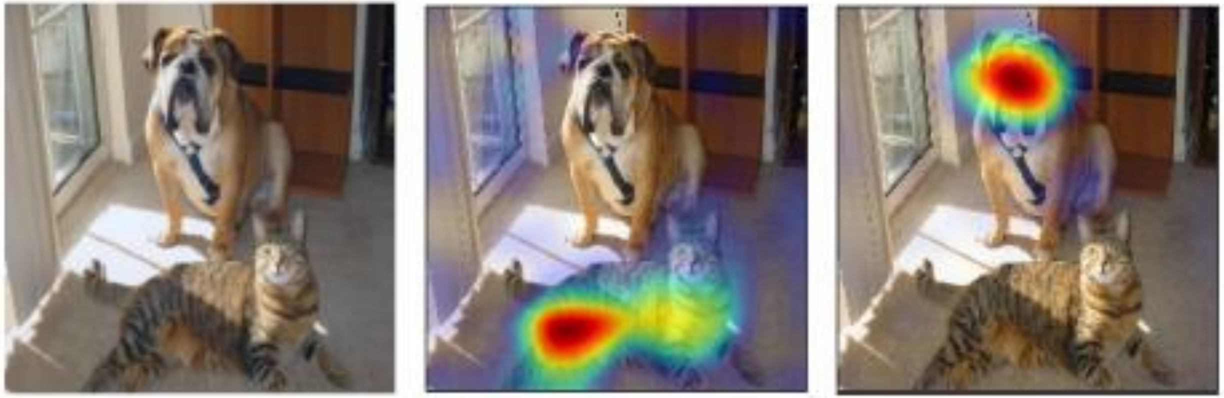




https://usmanr149.github.io/urmlblog/cnn/2020/05/01/Salincy-Maps.html
https://medium.com/@bijil.subhash/explainable-ai-saliency-maps-89098e230100
https://www.youtube.com/watch?v=ZIGff_iGlVY

Class Activation Map

CAM highlights the regions in an image that are most important for a CNN to make a specific classification decision.



Original Image Predicted as cat Predicted as dog

- Training a CNN several filters are learned for convolution operations and each filter application results in a feature map. Filters close to **the input layer of a CNN are often detecting low-level features**, such as edges or lines. The deeper you go, the more are those low-level features combined into **higher level features**.

Class Activation Map

CAM highlights the regions in an image that are most important for a CNN to make a specific classification decision.

1. Train a CNN

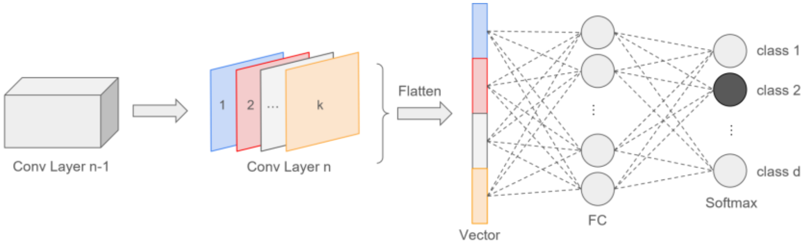
- Train a CNN for image classification.
- Ensure the final convolutional layer is followed by a global pooling layer (e.g., Global Average Pooling (GAP)) and a fully connected layer for classification.

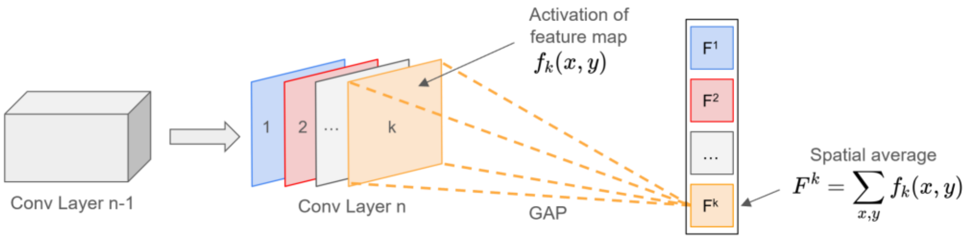
2. Identify the feature map Layer for CAM

- Select the last convolutional layer of the trained model, as it contains spatial information about the image.

3. Extract Feature Maps

- Pass the input image through the model to obtain the feature maps from the selected convolutional layer.





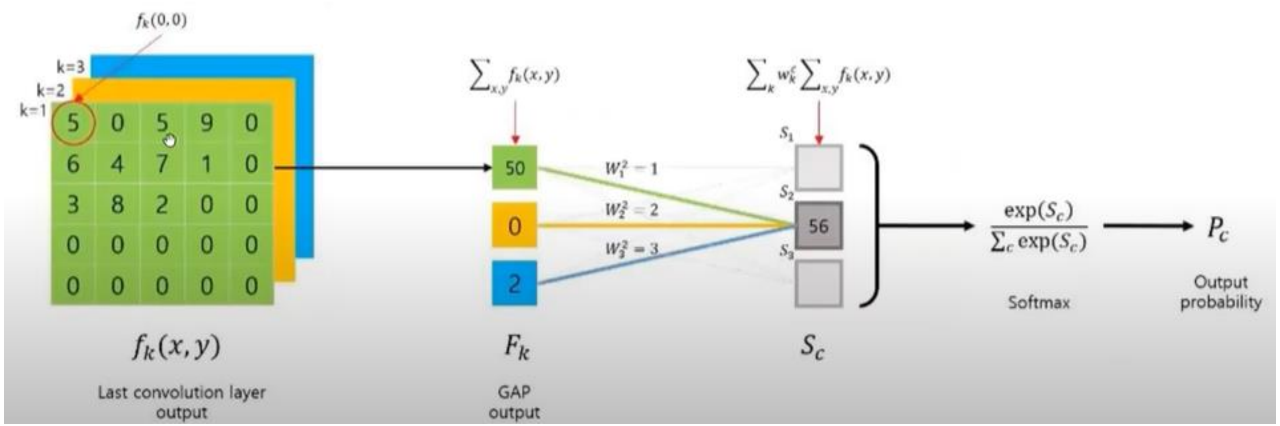
Class Activation Map

4. Access the Class-Specific Weights

- Retrieve the weights from the fully connected layer associated with the class of interest. These weights indicate the importance of each feature map channel for the specific class.

5. Compute the CAM

- Multiply each channel of the feature maps by its corresponding weight from the fully connected layer.
- Sum the weighted feature maps across all channels to generate a single activation map.
- Optionally, apply a ReLU to focus only on positive contributions.



F^k : GAP output of Kth feature map, $\sum_{x,y} f_k(x,y)$

w_k^c : the importance of F^k for class c

$$S_c = \sum_k w_k^c \sum_{x,y} f_k(x,y) = \sum_{x,y} \sum_k w_k^c f_k(x,y).$$

CAM

6. Upsample the CAM

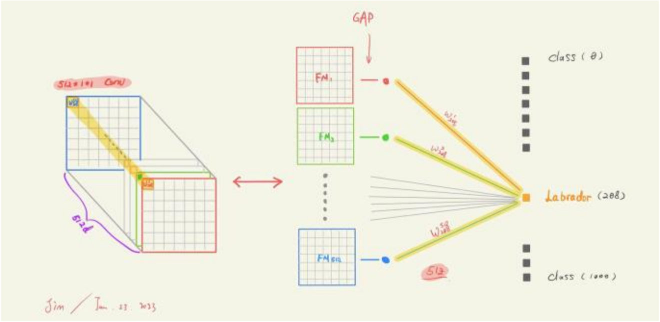
- Resize the CAM to match the dimensions of the original input image using interpolation (e.g., bilinear interpolation).

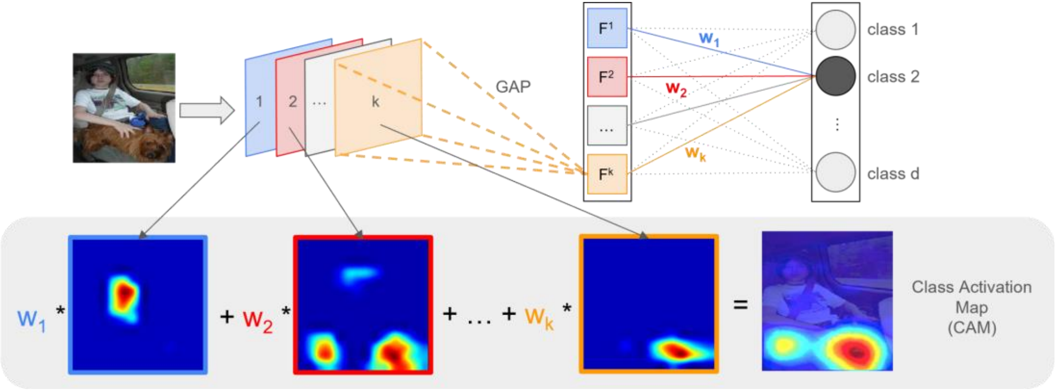
7. Overlay the CAM on the Image

- Normalize the CAM values to a range of 0 to 1.
- Apply a color map to the CAM and overlay it on the original image for visualization.

8. Interpret the Results

- Analyze the highlighted regions to understand the areas of the image that the model focused on for the classification.

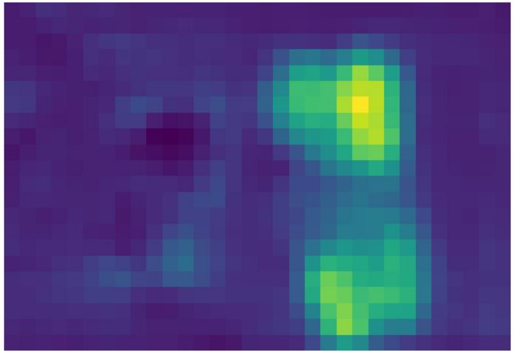




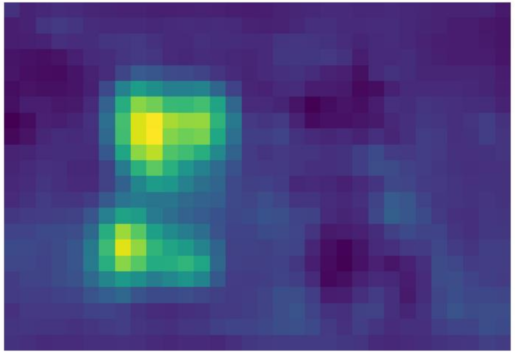
Zhou et. al Learning Deep Features for Discriminative Localization, CVPR 2016

https://github.com/JimEverest/CAM

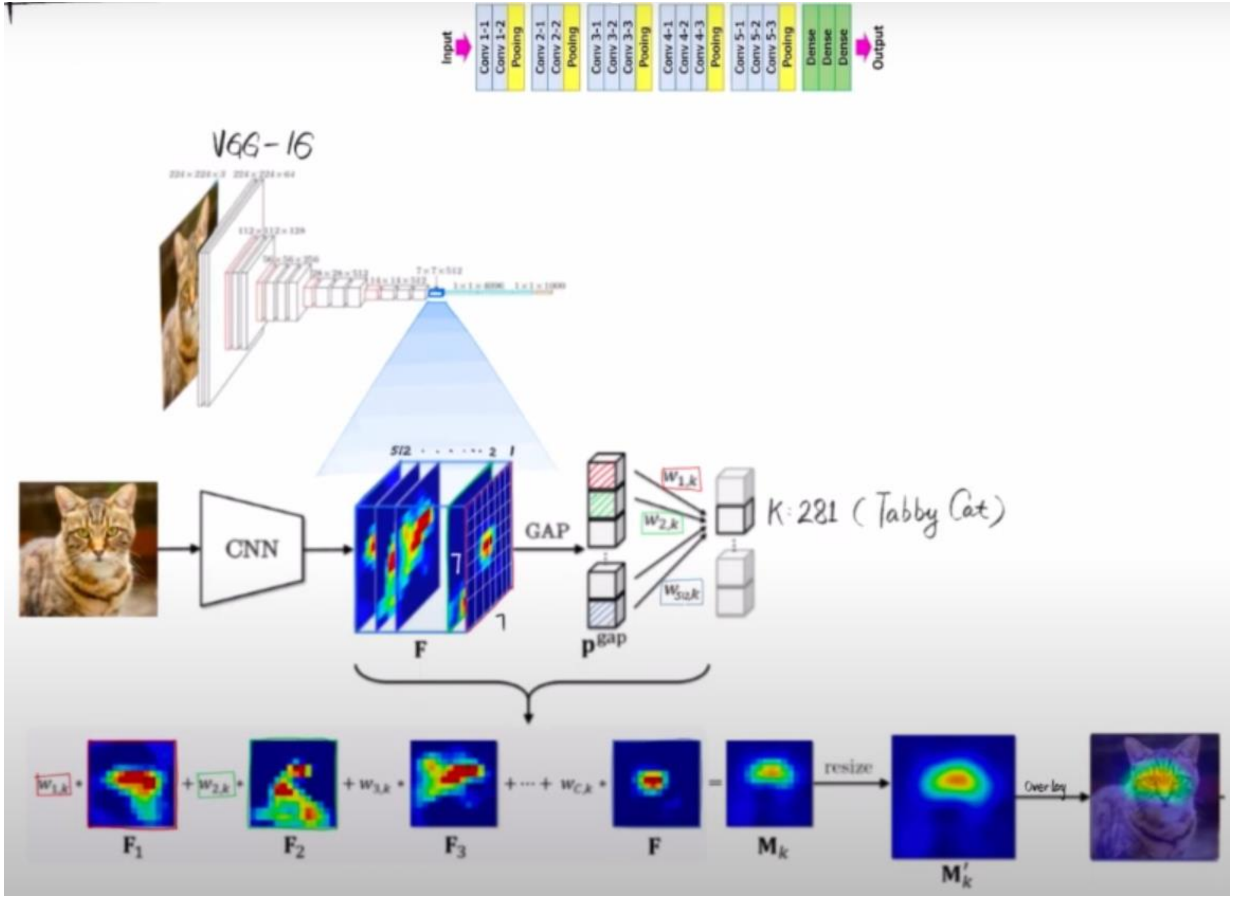
CAM Model Implementation






Recap CAM



Input
Conv 1-1
Conv 1-2
Pooling
Conv 2-1
Conv 2-2
Pooling
Conv 3-1
Conv 3-2
Conv 3-3
Pooling
Conv 4-1
Conv 4-2
Conv 4-3
Pooling
Conv 5-1
Conv 5-2
Conv 5-3
Pooling
Dense
Dense
Dense
Output
VGG-16
512 · · · · · · 2 1
CNN
GAP
7
7
F
$\mathbf{p}^{gap}$
$w_{1,k}$
$w_{2,k}$
$w_{512,k}$
K: 281 (Tabby Cat)
$w_{1,k}$ · + $w_{2,k}$ · + $w_{3,k}$ · + ··· + $w_{C,k}$ · =
$\mathbf{F}_1$
$\mathbf{F}_2$
$\mathbf{F}_3$
F
$\mathbf{M}_k$
resize
$\mathbf{M}'_k$
Overlay

Grad-CAM: Gradient-Weighted Class Activation Map

- Grad-CAM (Gradient-weighted Class Activation Mapping) is a **model-specific** method, which provides local explanations for Deep Neural Networks.

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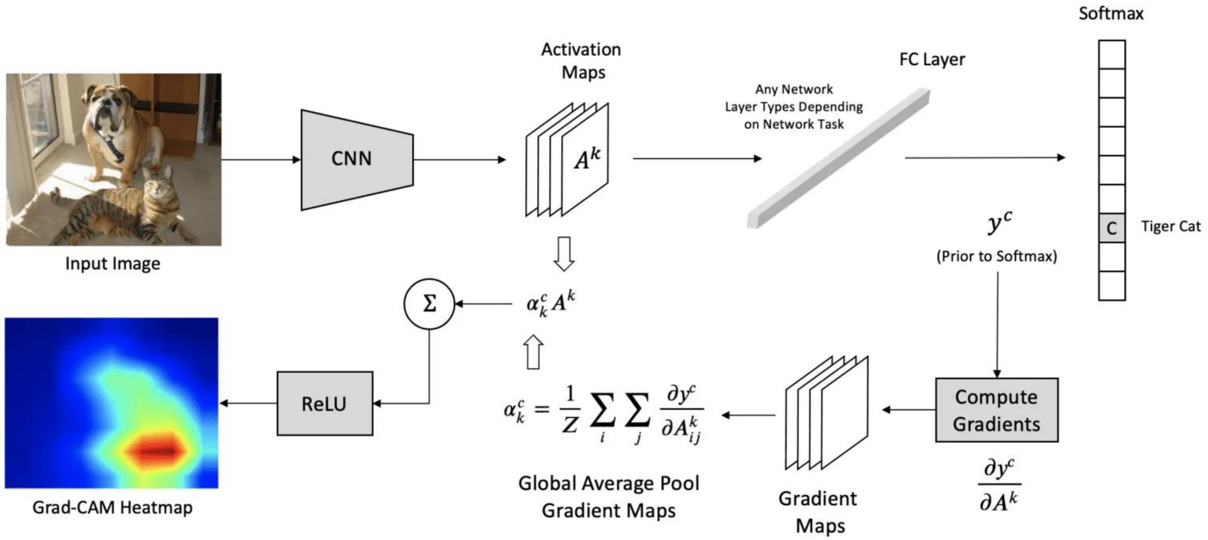
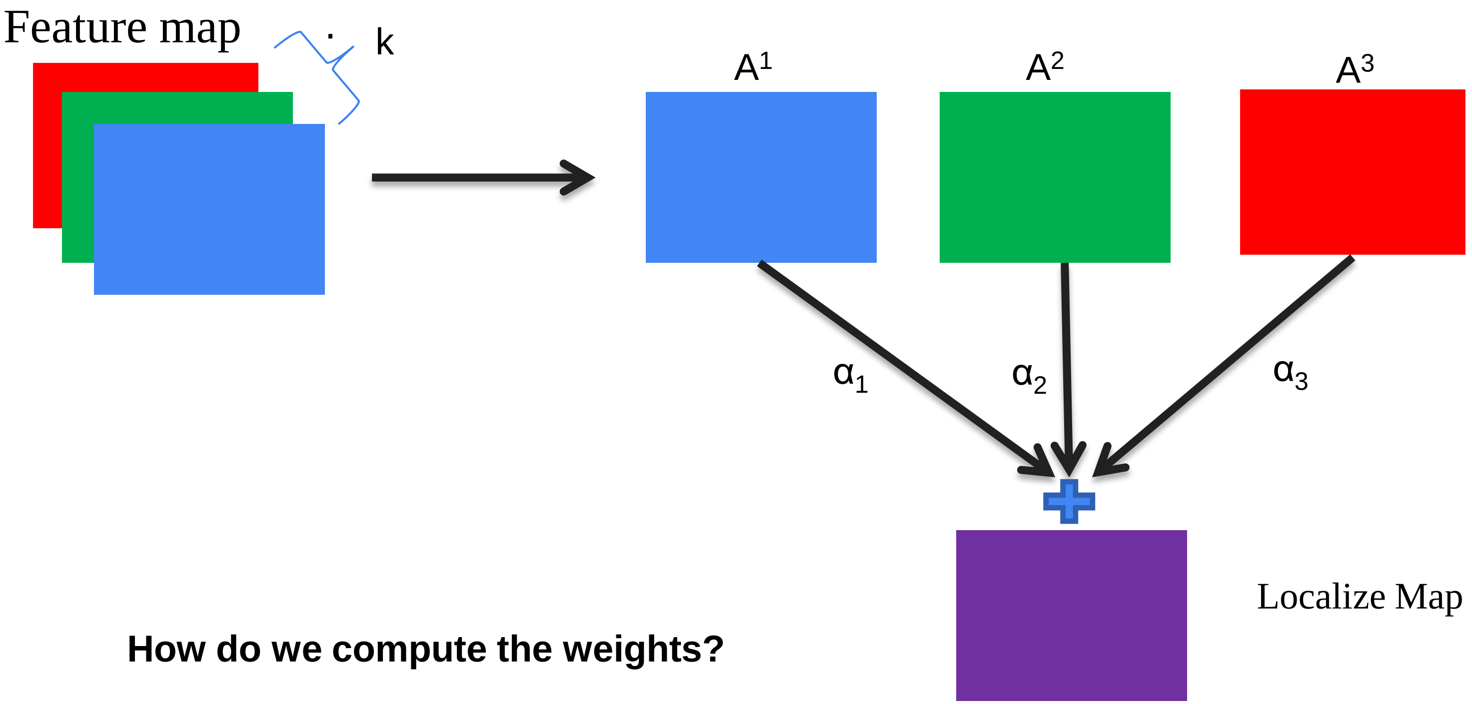


Selvaraju, et. al. Grad-cam: Visual explanations from deep networks via gradient-based localization. ICCV. 2017

Grad-CAM: Gradient-Weighted Class Activation Map

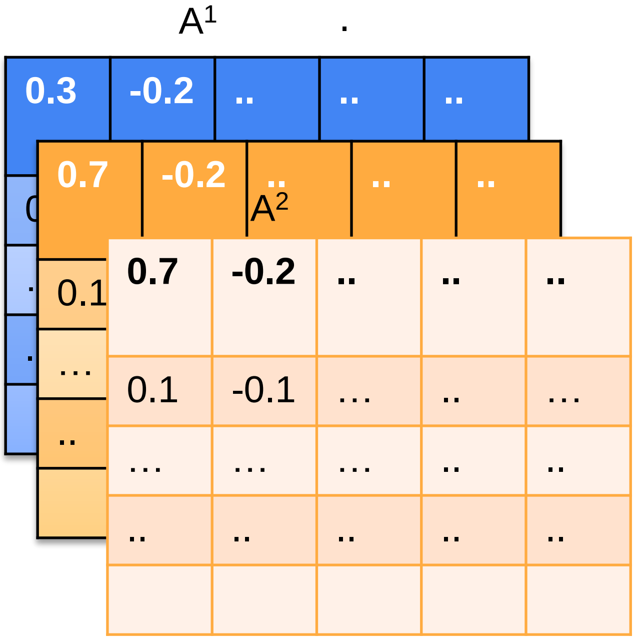


Localize Map (L) = $\alpha_k^c A^k$

How do we compute the weights?

Grad-CAM Steps

Step 1: Gradient Computation in GradCAM



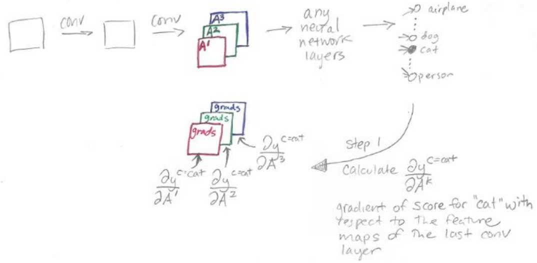
y output

$$\frac{\partial y^c}{\partial A^k}$$

Greatest positive influence

0.3	-0.2	..	..	..
-0.7	0.5	…	..	…
…	…	…	..	..
..	..	..	..	..

Greatest negative influence

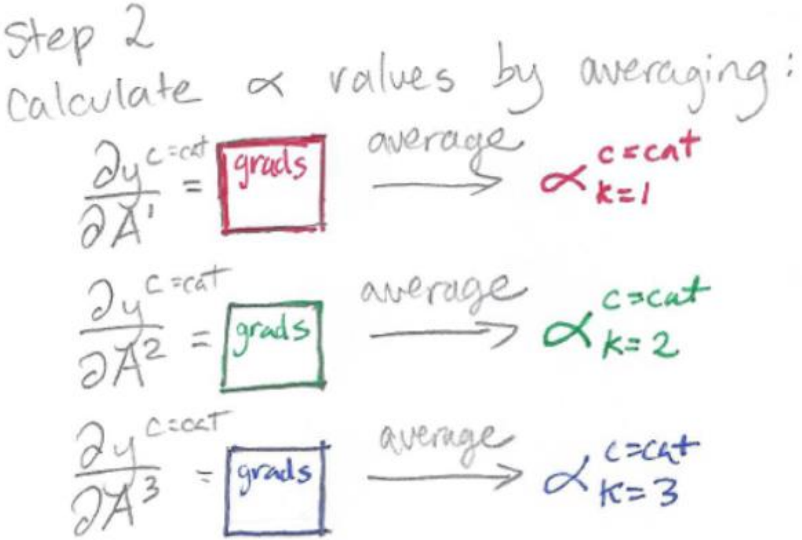


For the equation above, y_c, is a scalar (predicted score before the softmax computation), and A_k is a two-dimensional feature map. So, the gradient is also a two-dimensional map with the exact spatial dimensions as the feature maps, A_k

Grad-CAM Step

Step 2: Compute Alpha Values for GradCAM

Global average pool the gradient over the width dimension (index by i) and the height dimension (index by j) to obtain neuron importance weight $\alpha_k c$, where Z is the total number of pixels in each feature map.



$$\alpha_k^c = \overbrace{\frac{1}{Z}\sum_i \sum_j}^{\text{global average pooling}} \underbrace{\frac{\partial y^c}{\partial A_{ij}^k}}_{\text{gradients via backprop}}$$

Grad-CAM: Gradient-Weighted Class Activation Map

Step 3: Calculate Final Grad-CAM Heatmap

Perform a weighted combination of the feature map activations A_k where the weights are the $\alpha_k c$ *just calculated*

$$L^c_{\text{Grad-CAM}} = ReLU \underbrace{\left(\sum_k \alpha_k^c A^k \right)}_{\text{linear combination}}$$

Step 3

Calculate the final heatmap as a weighted combination of the feature maps:

$$\text{Grad-CAM}^{cat} = \alpha_1 A^1 + \alpha_2 A^2 + \alpha_3 A^3, \text{ i.e. } \sum_k \alpha_k^{cat} A^k$$

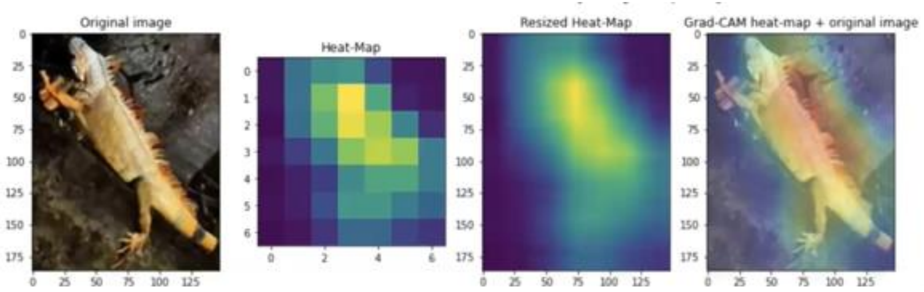
Note the similarity to the final CAM equation:

$$\text{CAM}^{cat} = W_1 A^1 + W_2 A^2 + W_3 A^3, \text{ i.e. } \sum_k W_k^{cat} A^k$$

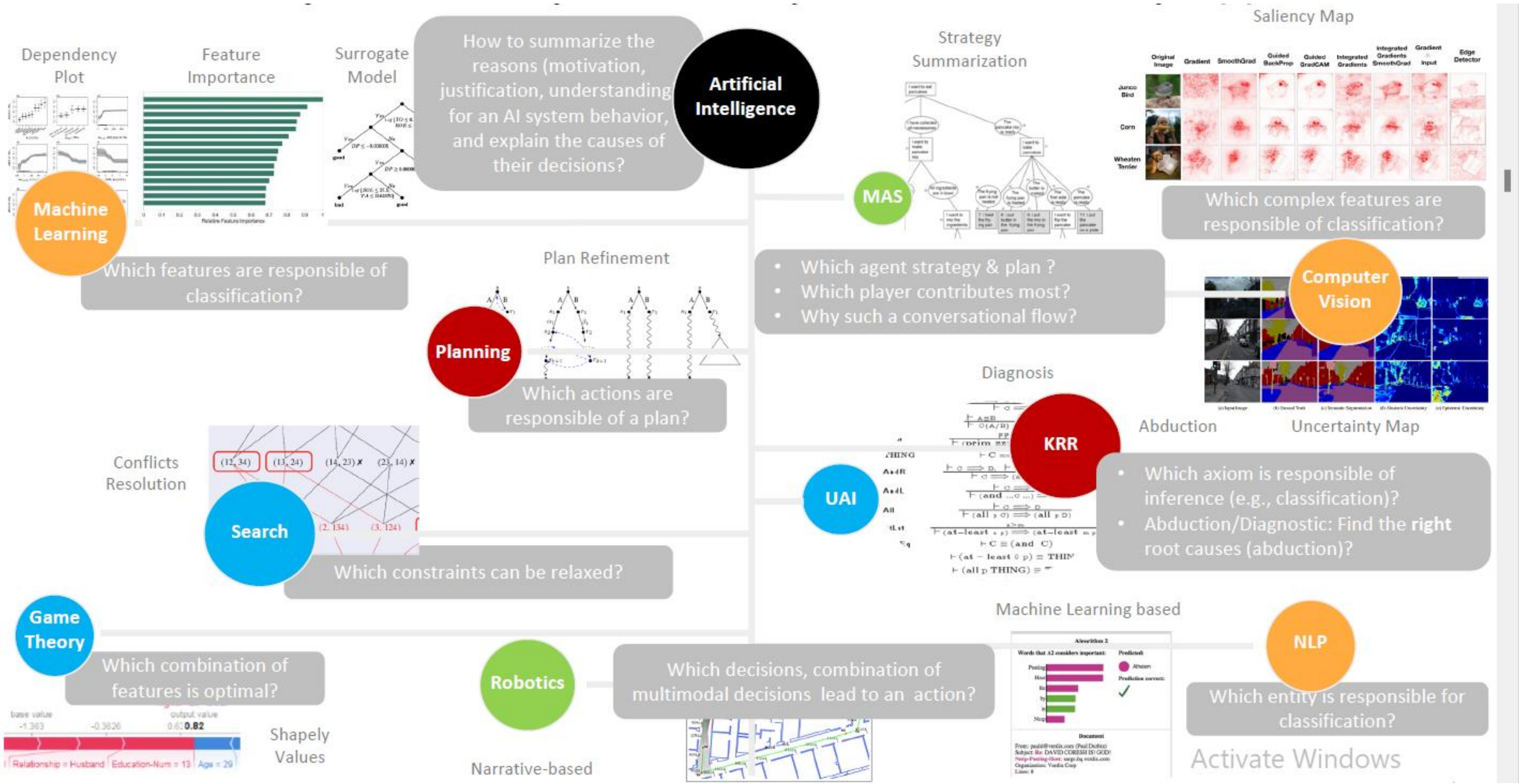
In Grad-CAM we also apply a ReLU:

$$\text{Grad-CAM}^{cat} = \text{ReLU}(\alpha_1 A^1 + \alpha_2 A^2 + \alpha_3 A^3)$$

Grad-CAM: Gradient-Weighted Class Activation Map



XAI: One Objective, Many 'AI's, Many Definitions, Many Approaches



"I'M A LITTLE SURPRISED. WITH SUCH EXTENSIVE EXPERIENCE IN PREDICTIVE ANALYSIS, YOU SHOULD'VE KNOWN WE WOULDN'T HIRE YOU."

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